

## VARENNES, QC, Canada

WMO: 711840

Lat: 45.7229N

Lon: 73.3765W

Elev: 18

StdP: 101.11

Time Zone: -5.00 (NAE)

Period: 94-16

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -23.5      | -21.1 | -28.7                         | 0.3 | -22.8 | -26.2 | 0.3 | -20.6 | 14.2                  | -1.6 | 12.5 | -5.1 | 3.9                   | 270  | 0.575 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.3                   | 30.3            | 21.8 | 28.6 | 20.9 | 27.2 | 20.1 | 23.5                | 28.1 | 22.4 | 26.7 | 21.5 | 25.4 | 4.8                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.9                            | 16.6 | 26.0 | 21.0 | 15.7 | 24.9 | 20.1 | 14.8 | 23.8 | 69.9          | 28.1 | 66.0 | 26.7 | 62.5 | 25.4 | 28.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 11.0          | 9.5  | 8.4 | DB                         | -27.4 | 32.7               | 3.4 | 1.5                                                | -29.8 | 33.8       | -31.8 | 34.7       | -33.7 | 35.5       | -36.2 | 36.5 | (4) |
| (5)               |      |     | WB                         | -27.6 | 25.5               | 3.3 | 1.2                                                | -30.0 | 26.3       | -32.0 | 27.0       | -33.9 | 27.7       | -36.3 | 28.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 6.7         | -9.5  | -8.1  | -2.2  | 5.9   | 13.6  | 18.6  | 20.6  | 19.6  | 15.5  | 8.6   | 2.1   | -4.8  | (6)  |  |
|                                                                        | DBStd       | 11.63       | 6.92  | 6.03  | 5.82  | 4.44  | 4.23  | 3.68  | 2.80  | 2.91  | 3.91  | 4.29  | 4.77  | 6.10  | (7)  |  |
|                                                                        | HDD10.0     | 2423        | 605   | 508   | 381   | 137   | 13    | 0     | 0     | 0     | 3     | 79    | 238   | 459   | (8)  |  |
|                                                                        | HDD18.3     | 4451        | 863   | 742   | 638   | 374   | 157   | 41    | 9     | 20    | 103   | 301   | 486   | 717   | (9)  |  |
|                                                                        | CDD10.0     | 1229        | 0     | 0     | 1     | 13    | 124   | 258   | 329   | 298   | 167   | 37    | 2     | 0     | (10) |  |
|                                                                        | CDD18.3     | 217         | 0     | 0     | 0     | 1     | 10    | 49    | 80    | 60    | 17    | 1     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 1761        | 0     | 0     | 1     | 9     | 137   | 424   | 613   | 444   | 130   | 4     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 425         | 0     | 0     | 0     | 2     | 33    | 118   | 148   | 96    | 28    | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 3.9         | 4.6   | 4.5   | 4.4   | 4.4   | 4.2   | 3.5   | 3.0   | 2.7   | 3.0   | 3.6   | 4.1   | 4.3   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1021        | 77    | 58    | 65    | 84    | 90    | 100   | 102   | 96    | 89    | 99    | 77    | 89    | (15) |  |
|                                                                        | PrecMax     | 1266        | 141   | 103   | 120   | 152   | 183   | 195   | 157   | 179   | 197   | 183   | 140   | 166   | (16) |  |
|                                                                        | PrecMin     | 824         | 27    | 22    | 18    | 17    | 43    | 45    | 53    | 39    | 25    | 30    | 12    | 27    | (17) |  |
|                                                                        | PrecStd     | 108         | 32    | 21    | 25    | 37    | 33    | 35    | 26    | 32    | 37    | 41    | 30    | 36    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 6.7   | 5.3   | 14.5  | 24.4  | 30.3  | 32.1  | 32.2  | 31.1  | 30.1  | 23.6  | 16.7  | 10.6  | (19) |  |
|                                                                        |             | MCWB        | 6.1   | 3.5   | 9.7   | 15.2  | 18.8  | 22.2  | 23.9  | 22.8  | 21.8  | 17.1  | 14.0  | 8.9   | (20) |  |
|                                                                        | 2%          | DB          | 4.1   | 3.5   | 9.9   | 19.5  | 27.0  | 29.8  | 29.9  | 29.1  | 26.7  | 20.4  | 13.6  | 7.0   | (21) |  |
|                                                                        |             | MCWB        | 2.8   | 2.1   | 6.3   | 11.5  | 16.5  | 20.6  | 22.2  | 21.7  | 20.2  | 15.3  | 11.5  | 5.8   | (22) |  |
|                                                                        | 5%          | DB          | 2.3   | 2.2   | 6.7   | 16.2  | 24.3  | 27.8  | 28.2  | 27.4  | 24.5  | 17.5  | 11.2  | 4.7   | (23) |  |
|                                                                        |             | MCWB        | 1.4   | 1.0   | 4.1   | 9.7   | 15.3  | 19.6  | 21.4  | 20.8  | 18.9  | 13.7  | 9.0   | 3.5   | (24) |  |
|                                                                        | 10%         | DB          | 0.6   | 1.0   | 4.8   | 13.3  | 21.7  | 25.8  | 26.7  | 25.9  | 22.4  | 15.2  | 9.0   | 2.7   | (25) |  |
|                                                                        |             | MCWB        | -0.2  | 0.1   | 2.8   | 8.0   | 14.4  | 18.5  | 20.5  | 20.2  | 17.7  | 12.0  | 6.9   | 1.7   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 5.9   | 3.9   | 11.0  | 16.5  | 21.2  | 23.7  | 25.2  | 24.6  | 23.2  | 18.5  | 14.9  | 9.2   | (27) |  |
|                                                                        |             | MCDB        | 6.4   | 4.6   | 13.0  | 22.7  | 27.6  | 30.0  | 30.9  | 29.2  | 27.8  | 21.8  | 16.1  | 10.6  | (28) |  |
|                                                                        | 2%          | WB          | 3.0   | 2.2   | 6.6   | 12.9  | 19.0  | 22.4  | 23.7  | 23.0  | 21.3  | 16.0  | 11.7  | 5.8   | (29) |  |
|                                                                        |             | MCDB        | 3.7   | 3.1   | 9.3   | 17.1  | 23.4  | 27.7  | 27.9  | 27.2  | 25.1  | 19.5  | 13.5  | 6.8   | (30) |  |
|                                                                        | 5%          | WB          | 1.4   | 1.2   | 4.5   | 10.4  | 17.2  | 21.1  | 22.6  | 21.9  | 19.9  | 14.2  | 9.2   | 3.5   | (31) |  |
|                                                                        |             | MCDB        | 2.2   | 2.0   | 6.3   | 14.8  | 22.0  | 25.7  | 26.6  | 25.7  | 23.3  | 16.8  | 10.8  | 4.5   | (32) |  |
|                                                                        | 10%         | WB          | -0.2  | 0.2   | 2.9   | 8.6   | 15.5  | 20.0  | 21.6  | 20.9  | 18.5  | 12.6  | 7.2   | 1.8   | (33) |  |
|                                                                        |             | MCDB        | 0.5   | 1.1   | 4.6   | 12.6  | 19.8  | 23.8  | 25.4  | 24.6  | 21.6  | 14.7  | 8.7   | 2.6   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.2   | 8.8   | 8.5   | 10.0  | 11.3  | 10.5  | 10.3  | 10.9  | 11.0  | 8.9   | 7.2   | 6.9   | (35) |  |
|                                                                        |             | MCDBR       | 8.7   | 8.1   | 9.9   | 15.6  | 15.5  | 13.5  | 12.1  | 12.5  | 13.1  | 12.4  | 10.5  | 7.5   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 8.5   | 7.5   | 7.3   | 9.1   | 7.6   | 6.5   | 6.1   | 6.5   | 7.3   | 7.8   | 8.6   | 6.9   | (37) |  |
|                                                                        |             | MCWBR       | 8.6   | 7.7   | 9.4   | 13.5  | 12.4  | 11.0  | 10.4  | 11.1  | 11.4  | 10.8  | 10.1  | 7.2   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.256 | 0.290 | 0.337 | 0.365 | 0.394 | 0.438 | 0.429 | 0.416 | 0.376 | 0.337 | 0.308 | 0.297 | (40) |  |
|                                                                        |             | taud        | 2.352 | 2.241 | 2.249 | 2.341 | 2.322 | 2.205 | 2.234 | 2.271 | 2.369 | 2.485 | 2.481 | 2.284 | (41) |  |
|                                                                        | Edn at Noon | Edn at Noon | 864   | 885   | 887   | 893   | 874   | 833   | 835   | 832   | 838   | 829   | 793   | 750   | (42) |  |
|                                                                        |             | Edh at Noon | 73    | 101   | 117   | 117   | 124   | 140   | 134   | 125   | 104   | 80    | 67    | 72    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.44  | 2.38  | 3.63  | 4.39  | 5.12  | 5.63  | 5.63  | 4.97  | 3.92  | 2.27  | 1.37  | 0.99  | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.17  | 0.25  | 0.30  | 0.50  | 0.49  | 0.42  | 0.44  | 0.23  | 0.32  | 0.20  | 0.18  | 0.08  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (46 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page